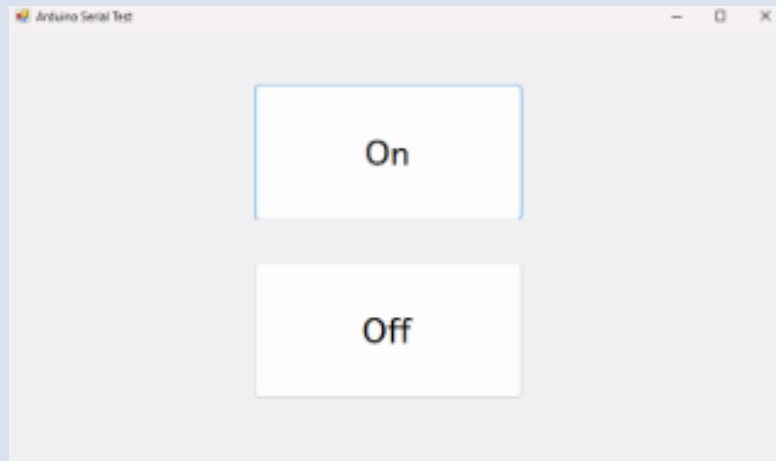


Advanced Task 3-2 LED Control with Serial Communication

Two-way UART communication between a C# host app and Arduino

<> Host PC App (C#)

- Design a GUI to control the LED
- When a button is pressed, send the corresponding value through the serial port



Arduino Firmware (.ino)

- Initialize Serial.begin(9600)
- Continuously listen for incoming serial strings
- Turn the LED on / off according to the data received from the PC app



Advanced Task 3-2 Development Hints



Break it into subtasks

This task is more complex than previous ones, but valuable for real project development. Split it into smaller pieces to complete it.



Choose the right project template

Use Visual Studio to build the PC host app. Investigate which project template fits this task yourself.



AI tools are allowed

You may use AI to help generate the sketch and source code — but you must understand it and be ready to explain it to the TA.



You don't need to master C#

You don't need to understand every function or command. Focus on understanding the code's structure and how it works.